

knowledge base can be called upon quickly and efficiently as per the requirement of any task – be it problem-solving, reasoning or anything else. Indeed, even the ability to solve one problem and use its logical inference to tackle another problem requires “knowledge” in the form of stored information about a certain way of doing one thing.

In the 1970s, there was a turnaround in the kinds of storage spaces available to computer scientists. The human brain tackles problems by using two aspects: knowledge and reasoning. As large amounts of memory started becoming available, researchers started dividing the artificial intelligence systems into two such halves as well.

Stanford University’s Edward Feigenbaum was at the forefront of this movement, developing what is the world’s first “expert system”. An expert system is essentially an intelligent agent that relies on two parts: the static inference engine and the dynamic knowledge database.

The inference engine is basically what Simon, Newell and McCarthy worked on – the ability of a computer to reason. For an expert system, it didn’t matter what kind of logical thinking it was based on, as long as the computer was able to reason. The real genius of this lied in its ability to use a knowledge database to store information as well as add to it over time. This way, the expert system was using reasoning capabilities with its pre-existing knowledge, making it more like a human brain than ever before. And over time, boffins developed the ability to add to the dynamic knowledge base, making it such that the AI could draw on previous logical inferences to further expand its knowledge base and thus increase the speed of reasoning.

Neat vs Scruffy

While the theories of Newell, Simon and McCarthy were being taken as gospel in most scientific communities, a new line of thinking came up in the mid-70s. Roger Schank, a cognitive scientist at Yale University, had a different interpretation of what was needed to solve the massive problem of coming up with an emulation of the human brain. His solution? Nothing.

Well, that’s putting it a bit harshly, but Schank’s premise was that the human mind is too vast and too complex to be reduced to a system that can be applied in any scenario homogenously. In his studies, he had found that different tasks required different parts of the human brain to kick in; and more importantly, people had different parts of the brain kicking in for the same task at different times or situations. So then how